

Enhancing Human Pose Estimation in Ancient Vase Paintings via Perceptually-grounded Style Transfer Learning

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Human pose estimation (HPE) is a central part of understanding the visual narration and body movements of characters depicted in artwork collections, such as Greek vase paintings. Unfortunately, existing HPE methods do not generalise well across domains resulting in poorly recognised poses. Therefore, we propose a two step approach: (1) adapting a dataset of natural images of known person and pose annotations to the style of Greek vase paintings by means of image style-transfer. We introduce a perceptually-grounded style transfer training to enforce perceptual consistency. Then, we fine-tune the base model with this newly created dataset. We show that using style-transfer learning significantly improves the SOTA performance on unlabelled data by more than 6% mean average precision (mAP) as well as mean average recall (mAR). (2) To improve the already strong results further, we created a small dataset (ClassArch) consisting of ancient Greek vase paintings from the 6–5th century BCE with person and pose annotations. We show that fine-tuning on this data with a style-transferred model improves the performance further. In a thorough ablation study, we give a targeted analysis of the influence of style intensities,

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revealing that the model learns generic domain styles. Additionally, we provide a pose-based image retrieval to demonstrate the effectiveness of our method. The code and pretrained models can be found at <https://github.com/angelvillar96/STLPose>.

CCS Concepts: • **Applied computing** → **Digital libraries and archives**; • **Computing methodologies** → **Visual content-based indexing and retrieval**; **Supervised learning**.

Additional Key Words and Phrases: Pose Estimation, Greek Vase Paintings, Style Transfer Learning, Digital Humanities

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1 INTRODUCTION

Human pose estimation (HPE) is highly challenging as it is difficult to have one method that can generalise across all domains. Estimating human pose involves localising each visible body-keypoint (Fig. 1c), however the state-of-the-art (SOTA) methods underperform when tested on different domains, for example ancient Greek vase paintings (Fig. 1). HPE is central to understanding the visual narration and body movements of the characters depicted in these paintings and the recent rapid digitisation of art collections has created an opportunity to use HPE as a tool to digitally examine such artworks. These digital copies are usually either photographic reproductions [37] or scans of existing archives [48]. In addition to the preservation of cultural heritage, these digital collections allow remote access of the invaluable artistic data to the general public. However, due to content complexity and large size, navigating within such collections is often daunting.

To address the challenge of efficiently analysing large digital collections, several computer vision and image analysis techniques have been used for applications, such as artist identification [28], object recognition [10, 19, 55], character recognition [34], artistic image classification [6, 8, 47] and pose-matching [27]. However, when these methods are evaluated on a different domain, they show sub-optimal performance, c.f. Fig. 1. Hence, an important challenge is to learn effective representations using little data. Human pose representation is one such example.

The understanding of visual narration in Greek vase paintings is one of the main objectives in the field of Classical Archaeology. In order to display the actions and situations of a narrative, as well as to characterise the protagonists, ancient Greek artists made use of a broad variety of often similar image elements [17]. Some of the key aspects of the narrative are illustrated by meaningful interactions and compositional relationships (*e. g.*, postures or gestures) between the characters displayed in the painting [36]. For example, the divine pursuit scene (Fig. 2, 1st & 2nd column) is a recurrent narrative in Greek vase paintings, often characterised by a character moving fast from left to right and reaching out with both his arms to catch a woman on her forearm or shoulder [49].

In this work, we propose to exploit these recurrent character interactions and postures in order to navigate semantically through collections of Greek vase paintings. We address image retrieval in such databases by measuring the similarity between character postures. Since ancient Greek artists made use of the postures to depict similar narratives, the retrieved images should display the same scene.

For human pose-based image retrieval in Greek vase paintings, we need a reliable human pose estimation algorithm. We propose a two-step approach: (1) First, we apply style-transfer [23] to the COCO [32] dataset to generate a synthetic annotated dataset with the style of Greek vase paintings and fine-tune the baseline person detection and pose estimation models on this dataset (Fig. 5 middle). (2) Second, we fine-tune these models on a



Fig. 1. Attic red-figure, *Zeus and Ganymede*, in ancient Greek vase paintings: (a) original image, (b) pose estimation by OpenPose [5], (c) our method.

newly generated dataset for Classical Archaeology (Fig. 5 bottom). We show that both steps improve the person detection and pose estimation tasks and thus the retrieval performance considerably.

In particular, our main contributions are:

- (1) We introduce the Styled-COCO-Persons (**SCP**) and the ClassArch (**CA**) datasets. *SCP* is a synthetic dataset, generated by applying style transfer, with different style-intensities, to the images from COCO (only ‘person’ class) to mimic the style of the **CA** dataset (Greek vase paintings). The **CA** dataset consists of 1783 images (characters) from 1000+ Greek vase paintings along with pose keypoint annotations.
- (2) We show that by just using styles of the **CA** dataset on real images, one can improve the task of human pose estimation in Greek vase paintings without requiring any annotations. We also show that fine-tuning this model with the small **CA** dataset modestly enhances the performance compared to direct transfer learning. Moreover, styled-tuned models outperform state-of-the-art fine-tuned methods on the **SCP** and **CA** datasets.
- (3) We introduce a perceptual loss for style-transfer and show that this is beneficial for both person detection and pose estimation.
- (4) Additionally, we show that our styled transfer learning based pipeline is also beneficial for retrieving and discovering similar images based on poses of the character in narratives from ancient Greek vase paintings.

2 RELATED WORK

The task of representing human poses has been studied since the early days of computer vision [39]. However, its importance is much older. *Pathosformeln* [2], the iconic study of basic constructs (units) of body language is one of the firsts to view the body gesture (or posture) also as a way of voicing inner emotions. Body movement’s depiction is essential and central to the historian [3, 4], since it gives a way to recognise the inner emotions or expressions of the character. Impett *et al.* [24] studies it with a geometrical construct by operationalisation of



Fig. 2. (1st column) Divine pursuit scene in ancient Greek vase paintings. The central character, a *winged persecutor*, is depicted with a similar pose, *i. e.*, arms extended towards the right (observer viewpoint) and legs with large strides. (2nd column) Leading the bride scene, with the central character *bride* depicted with similar poses with her left hand extended forward (observer viewpoint) held by the groom. (3rd column) Abduction scene, where character on the left is abducting the character on the right. (4th & 5th columns) Wrestling in Agonal and Mythological contexts between two main characters.

body movements, which could be considered a way of Pathosformeln. McNiven *et al.* [36] also used human poses as the basis to study the interactions between characters in ancient vase paintings.

Since several years, Convolutional Neural Networks (CNNs) have been dominating computer vision tasks, and HPE is not an exception. After *DeepPose* [53], methods followed that improved HPE by using CNN cascades [54] and graphical models [52]. Two major types of approaches are popular with HPE: *bottom-up* and *top-down*.

Bottom-up pose estimation [5, 26, 42] directly estimate the location of all keypoints and assemble them into pose skeletons for all people in the image simultaneously. They use CNNs like ResNet [20] and DenseNet [22] as backbones to predict keypoints and optimisation-based matching techniques like *DeepCut* [42] and *DeeperCut* [26] for combining the keypoints into poses. Cao *et al.* [5] introduced *Part Affinity Fields* (PAFs) which is able to estimate poses in real-time by solving a bi-partite graph matching problem, as a way to solve the optimisation problem of aggregating poses. Bottom-up techniques' lack of structural information leads to many false positives and often being outperformed by top-down pose estimation approaches.

Top-down pose estimation [9, 13, 41, 56] approach HPE in two steps (Fig. 5, *first row*). The first step addresses the problem of detecting all person instances in the image, whereas the second step aims at predicting the body-keypoints for each of the detected person. For person detection, a specific CNN, *e. g.*, from the R-CNN family [15, 16, 43] is normally used. The second step involves using a single-person pose estimation model to process each of the person instances independently. Tompson *et al.* [52] first proposed the use of a CNN with multi-resolution receptive fields for the task of body joint localisation. More recent methods, however, focus on refinement techniques, such as Iterative Error Feedback [7], Stacked-hour-glass networks [40] and PoseFix [38]. Multi-scale approach by Sun *et al.* [51] uses a novel architecture to maintain high-resolution representations through the whole estimation process by repeated multi-scale feature fusions. These SOTA methods have improved the performance on their respective benchmark datasets, however, they fail to generalise to domains like Greek vase paintings.

Domain adaptation based approaches mainly aim at bringing the distributions of source domain closer to that of the target, so that a single model can be used for both domains. Methods using domain adaptation via style-transfer mainly transfer the style of target to the source while training online as a way of domain adaption, using style loss [44] or even adapting the domain progressively [25]. Some methods also use feature level

Table 1. **Datasets** used in our experiments. For COCO-Persons (*CP*), we use images from the person category of COCO [32]. Styled-COCO-Persons (*SCP*) is generated by using *CP* images as *content* and various splits of ClassArch (*CA*) images as *styles* (*Images*: images, *Persons*: person bounding boxes, *Poses*: pose annotations).

Datasets→	CP & SCP			CA		
Split→	Train	Val	Total	Train	Val	Total
Images	64115	2693	66808	1210	303	1513
Persons	257252	10777	268029	2098	531	2629
Poses	149813	6352	156165	1425	303	1728

alignment for aligning the two domains [31], and others enforce it via self-similarity and domain-dissimilarity loss [12]. Recently, various methods have focused on reducing data bias in order to enhance the transfer-ability of features in task-specific layers [33, 35]. [45] and [50] proposed unsupervised domain adaptation techniques, using minimum entropy consensus and co-variance of source and target features respectively.

A few works using generative models have also been proposed. [46] proposed a method for optimising bi-directional image transformations and using class consistency loss, while [21] proposed a cycle GAN that adapts representations to combine feature-level and pixel level while enforcing structural (cycle loss) and semantic consistency (task-specific). A work closely related to ours [27], highlights the importance of using poses for artwork discovery and its subsequent analysis. Their study, however, is based on artworks with some case study images that are relatively easy for SOTA methods, such as OpenPose [5] to estimate the poses. One of the first work done on Greek Vase paintings was by Crowley *et al.* [11] to generate a correspondence between descriptions and unknown regions in the images of vase paintings. In their work, they describe the challenges of working with Greek vases and annotate the images automatically. On a similar note, estimating poses for characters in ancient Greek Vase paintings presents a completely different challenge, since OpenPose fails very often (Fig. 1).

Our work’s focus, instead, is on using the style transfer to generate a synthetic dataset from already existing labelled dataset like COCO to improve the pose estimation on unlabelled data like ancient Greek vases, by enforcing a pre-computed perceptual consistency loss.

3 DATASETS

In order to train deep networks in a supervised fashion, it is very important to have a high-quality annotated dataset. Hence we work with 3 main datasets *viz.* **COCO-Persons (CP)** with images of only the ‘person’ category, its corresponding styled counterpart called **Styled-COCO-Persons (SCP)**, and we also introduce our own annotated dataset called **ClassArch (CA)**. Each dataset is labelled with person bounding boxes and their corresponding body pose keypoints, details shown in Tab. 1. We focus on these datasets for training and evaluation of our models.

(a) COCO-Persons (CP) The Common Objects in COntext dataset (COCO) [32] was specifically designed for the detection and segmentation of objects in their natural context. COCO has 328K images with over 2.5M labelled instances divided into 91 semantic object categories (*e. g.*, car, person, dog, banana, *etc.*). We only consider images that include “person” instances, along with their corresponding bounding boxes and pose-keypoints. We call this split as **COCO - Persons (CP)**. This split is taken across the training and validation sets only since the labels for the test set are not publicly available. Consequently, we use the validation set for testing our models. Tab. 1 shows the exact splits for the dataset in terms of images, persons and pose-keypoints. Figs. 4a & 4b illustrate some samples of *CP* dataset.

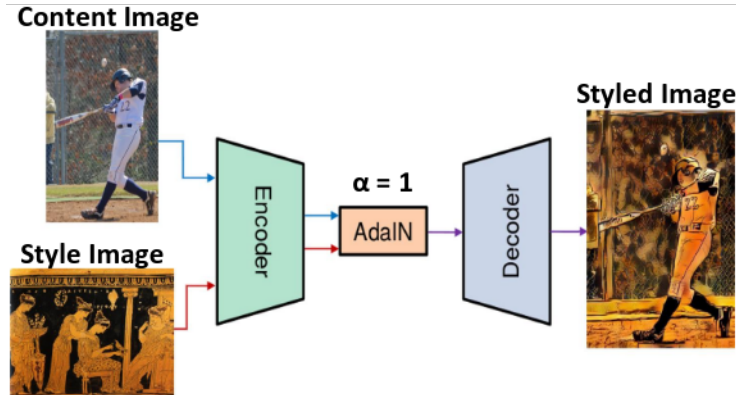


Fig. 3. Style transfer using *AdaIN* [23] with full style intensity ($\alpha = 1$). *AdaIN* adjusts the first and second order moments of the ‘Content Image’ to match those of the ‘Style Image’. A ‘Styled Image’ (style-transferred) is generated with the semantic content of the ‘content image’ and style of the ‘Style Image’.

(b) ClassArch (CA) We introduce a challenging dataset from the domain of Classical Archaeology, called **ClassArch** (CA) dataset. We chose five different recurrent narratives, viz. ‘Pursuits’, ‘Leading of the Bride’, ‘Abductions’, and ‘Wrestling’ in *Agonal* and *Mythological* contexts, taken from the period between the 6th and 5th century BCE. Pose-based analysis of such paintings is of critical importance for Classical Archaeology as discussed in Sec. 1. Each of the narratives in CA has its own set of characters, which appear recurrently and are depicted with similar features and in almost identical poses. Figs. 4e & 4f illustrate some images and their corresponding person bounding boxes and person keypoints of the CA dataset. Fig. 2 displays two examples from the ‘Pursuit’ (1st & 2nd column) and ‘Leading the bride’ (bottom row) narratives. In both scenes, the main characters (‘persecutor’/‘fleeing’ & ‘bride’) are depicted with similar posture in every image. CA has different sets of labels associated with it. There are 1513 images, with 2629 person annotations and 1728 pose annotations. More detailed splits are shown in Tab. 1.

(c) Styled-COCO-Persons (SCP) The images in CP significantly differ in semantic content and style from the ancient Greek vase paintings, c.f. Fig. 4a vs. Fig. 4e). To bridge this domain gap between CP and CA, we use style transfer to adapt the style of the CP dataset to vase paintings.

Style transfer algorithms render a synthetic image that combines the semantic information from one input (denoted as *content image*) with the texture from the user-defined *style image* [14]. We apply an efficient and fast style transfer technique using adaptive instance normalisation (*AdaIN*) [23] to create SCP, a synthetic dataset that combines the semantic content of the CP with the style of CA. Fig. 3 illustrates the style transfer procedure. We can visually observe that images of Figs. 4c & 4d are more closer in styles with Fig. 4e than Fig. 4a, i. e., SCP is closer in style with CA, than CP is with CA. The SCP dataset will be released along with the code.

Alpha (α) and Style-Sets Huang *et al.* [23] suggest a content-style trade-off technique to control the intensity of style transferred to the content image using $\alpha \in [0, 1]$. Based on this, we generate 2 groups of SCP. First with $\alpha = 0.5$, meaning that we only transfer half of the style intensity to the content images; and a second one in which α is chosen randomly from the uniform distribution $U[0, 1]$. The second group contains images across the whole spectrum, from no style ($\alpha = 0$) to full style ($\alpha = 1$).

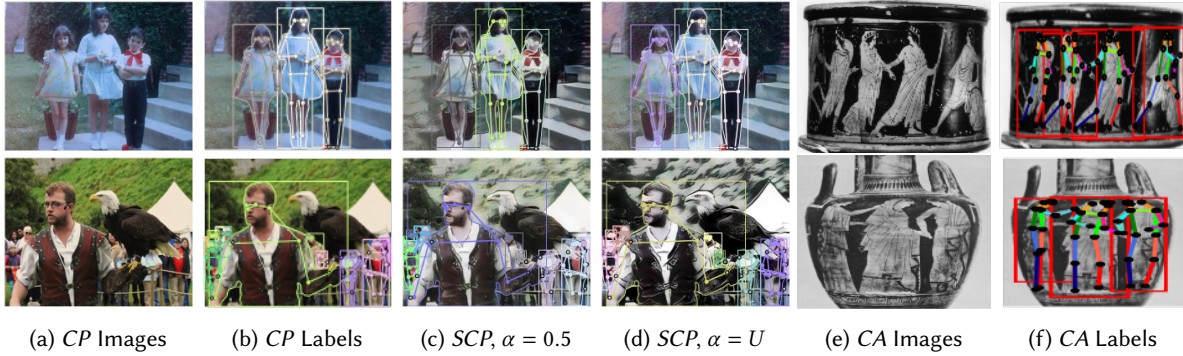


Fig. 4. **Dataset Samples:**(a) Images & (b) Labels of *CP* dataset; (c) & (d) are samples from the *SCP* dataset with $\alpha = 0.5$ and $\alpha = U$ respectively; (e) shows images with (f) the corresponding labels of our *CA* dataset. Each labelled example shows the corresponding person bounding boxes and their pose keypoints.

Additionally, we generate two more *SCP* dataset variations with the method described above using a different dataset of style images. We name this dataset as **Red-Black** figures (100 in total) or just **RB**, they are similar in style to the *CA* dataset but do not have any labels. Our hypothesis is that the model should be able to learn the styles and not the content of style images. In the end, we have four groups of *SCP* dataset with two different combinations of α with the two style-sets *RB* and *CA*.

Open Source Images (OSI) We will publicly release the *CP* and *SCP* datasets + annotations. 71 sample OSI links from the *CA* dataset are in the supplementary material, the rest can be downloaded from Beazley Archive Pottery Database¹ for research purposes. We'll release the permanent links to those images, along with the pose and person annotations.

4 PROPOSED METHOD

In this section, we first present our proposed style-based transfer learning approach to enhance pose estimation. We then briefly explain our models that were trained and evaluated for all datasets mentioned in Sec. 3. Lastly, we propose a perceptual loss as a regularizer to improve the estimation of perceptually similar poses with different styles.

4.1 Pose Estimation Approach

We take a top-down approach to pose-estimation, which is divided into two stages. Fig. 5 details the two stages of the top-down approach. The first stage (A/A^*) detects all the persons in an image and then estimates the keypoints (B/B^*) for each person instance, then creating the poses for each instance by pose-parsing (C/C^*). The models without * are trained on styled datasets while the ones with * are fine-tuned on the *CA* dataset. We use Faster-RCNN [43] as our person detector that was trained on the COCO [32] dataset. Top-down pose estimation approaches dominate the COCO keypoint detection challenge in the past few years, and several use the HRNet [51] as their backbones. Hence, we chose HRNet-W32 (henceforth denoted as *HRNet*) as our pose estimation model.

2-Step Training Approach. We adopt a 2 step approach to enhancing pose estimation, as shown in Fig. 5. In the first step (Fig. 5, *second row*), we train our detector and pose estimation model on styled data (different groups of

¹<https://www.beazley.ox.ac.uk/pottery/default.htm>

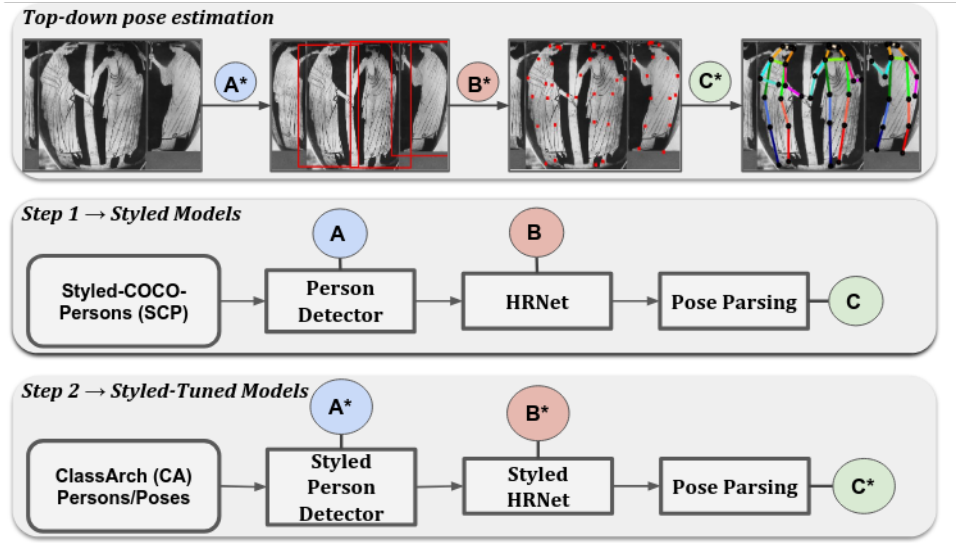


Fig. 5. (first row, *Top-down pose estimation*) - (A*) styled person detector detects all instances, (B*) for which the body joint locations are predicted using a person keypoint detector, (C*) The pose skeletons are assembled by connecting the detected keypoints for each person. **2 Step Training Approach: Step 1** (second row, *Styled Models*) Person Detector trained on SCP persons data, and HRNet on SCP poses data; **Step 2** (third row, *Styled-Tuned Models*) Styled Person Detector from second row is fine-tuned on CA persons data, and Styled HRNet is fine-tuned on CA pose data.

SCP) to generate *styled* models (A and B in Fig. 5). In this step, the *styled* models at the end of their training are expected to learn the styles of the target data, while trying to maintain the performance on the original task.

In the second step (Fig. 5, *third row*), we fine-tune these *styled* models on our CA data. During this step, the models that have learned the styles in the first step, now focus on improving their performance for the target dataset of CA. The final detector (A*) and pose estimator (B*) models are initialized with the A and B models from the first step and then fine-tuned on SCP persons and poses data.

We report all our experiments using four kinds of models for both tasks, person detection and pose estimation.

1. Baseline models are SOTA models. In case of the person detector, we drop the heads for all the classes except one, and fine-tune it on the ‘person’ class, further denoted as our *baseline* model.

2. Tuned models are SOTA models fine-tuned on the CA dataset. For the detector, we drop all the heads except one (similar as for baseline model) in Faster-RCNN and fine-tune it on persons data of the CA dataset. Likewise, for pose estimation, we take the SOTA HRNet and fine-tune it on pose data of CA.

3. Styled models are SOTA models trained on a particular group of the SCP dataset. As explained in Sec. 3, there are 4 different groups of SCP dataset. Depending on the values of α and *style-set* (RB or CA), the *Styled* models are trained on that particular group, for the detector as well as the pose estimator. Accordingly, there are 4 different *Styled* models for each of the detector and the pose estimator.

4. Styled-Tuned (Sd→Td) models are *Styled* models ((3) above) fine-tuned on CA dataset. Accordingly, for the detector, Sd→Td model is a *Styled* Faster-RCNN model fine-tuned on CA persons data. Similarly, for the pose

estimator, $Sd \rightarrow Td$ model is a *Styled* HRNet model fine-tuned on *CA* poses data. Hence, depending on the group of the *Styled* model, there is an equivalent $Sd \rightarrow Td$ model.

4.2 Enforcing Perceptual Similarity

While training the *styled* models, the network is fed with styled data. The advantage of doing this is to allow the model to expand its capacity to recognise perceptually similar persons/poses with different styles. In order to achieve content consistency in the perceptual space, we enforce a pre-computed perceptual loss [29] while training, in addition to the regular loss. The model is penalised if it is not able to maintain perceptual consistency.

Let's denote the task loss by L_T , where $L_T = L_{det}$ is for detector models, and $L_T = L_{pose}$ for pose models. We adopt two flavours of the combined loss, each for the detector as well as pose. In the first flavour (L_{comb1}), we adaptively weigh the perceptual loss ($L_{percept}$) with the corresponding detector or the pose loss, as shown in Eq. 1a. While, in the second flavour (L_{comb2}), we weigh each loss term with optimal values of λ_1 and λ_2 chosen using hyperparameter optimisation, as shown in Eq. 1b.

$$L_{comb1} = L_T + L_T * L_{percept} \quad (1a)$$

$$L_{comb2} = \lambda_1 * L_T + \lambda_2 * L_{percept} \quad (1b)$$

5 EXPERIMENTS AND ANALYSIS

The exact number of images, person bounding boxes and pose annotations, along with the corresponding train/val splits used for our experiments are mentioned in Tab. 1. In this section, we describe the evaluation protocol to train our detector and pose estimator. We also present the experimental results and discuss our findings.

5.1 Training Setup

In general, we use the standard parameters of the SOTA models and make adjustments to suit our experimental needs. For person detection (Faster-RCNN), we use an initial learning rate (lr_{init}) of 0.0001 with a scheduler $lr_{scheduler}$ on plateau (3 epochs) which reduces the lr by a factor of 0.33. We use Adam [30] with its default parameters and a batch-size (bs) of 8. Standard multi-task loss metric, a combination of log loss and regression loss ($L_{det} = L_{CLS} + L_{Reg}$), is used in our experiments for the detector. We train for 25 epochs on the *CP* dataset and 30 each on the *SCP* and *CA* datasets. However, we found that our models usually converge between 8–12 epochs. Similar to the detector, we use Adam with its default values for pose models (HRNet) and $bs=64$. With an lr_{init} of 0.01 and a $lr_{scheduler}$ on plateau (3 epochs) which reduces the lr by a factor of 0.1. We train all pose models for 100 epochs. Akin to the original HRNet [51], we also use the same configs (augmentations, image size) for fair comparison. Like HRNet, Object Keypoint Similarity (OKS) is used as an evaluation metric in our experiments as a simple Euclidean distance ($L_{pose} = L_{MSE}$) for pose estimation.

In both cases, person detection and pose estimation, we report the mean Average Precision (mAP) as well as the corresponding mean Average Recall (mAR).

5.2 Experiments

We compare different models in separate tables to give a clear understanding of our methods. As described in Sec. 3, **Style-set** (SS) represents two datasets *CA* and *RB* and **alpha** (α) represents the amount style transferred in the *SCP* dataset from no style ($\alpha = 0$) to full style ($\alpha = 1$). Tab. 2 (Results A) compares the *baseline* models with *styled* models for detector as well as pose. Similarly, Tab. 3 (Results B) compares the *tuned* with *styled-tuned*. Tab. 4 (Results C) shows the influence of using different data quantities to fine-tune our models, where as Tab. 5 shows the advantage of using perceptual loss.

Table 2. **Results A:** Comparing *baseline* model and *styled* models with different combinations of α (stylization factor) and SS (**RB** or **CA**), (a) for pose estimator and (b) person detector. $\alpha = 0.5$ or $\alpha = U$ (randomly sampled from a uniform distribution: $U = \text{uniform}(0, 1)$). All values in terms of *mAP*, except CA_{mAR} (*mAR*). The (+/-) is in reference to the corresponding baselines.

Model	CP	SCP	CA	CA_{mAR}	α	SS
Baseline	76.5	46.2	24.7	30.9	-	-
Styled	73.4 (-3.1)	54.4 (+8.2)	29.7 (+5.0)	36.0	0.5	RB
	74.0 (-2.5)	53.7 (+7.5)	30.9 (+6.2)	37.7	U	RB
	74.0 (-2.5)	53.8 (+7.6)	30.6 (+5.9)	36.9	0.5	CA
	74.3 (-2.2)	53.5 (+7.3)	32.3 (+7.6)	39.0	U	CA
(a) Pose Estimation						
Model	CP	SCP	CA	CA_{mAR}	α	SS
Baseline	39.4	24.2	10.4	9.8	-	-
Styled	37.5 (-1.9)	33.4 (+9.2)	7.6 (-2.2)	9.8	0.5	RB
	36.9 (-2.5)	32.1 (+7.9)	6.5 (-3.3)	8.5	U	RB
	37.7 (-1.7)	33.2 (+9.0)	8.2 (-1.6)	10.0	0.5	CA
	37.0 (-2.4)	32.6 (+8.4)	6.5 (-3.3)	8.6	U	CA
(b) Person Detection						

Baseline vs Styled models. (Results A, Tab. 2) It is important to understand the impact of styles on the main task for detection and pose estimation. We study the impact of style-transfer by comparing *baseline* and *styled* models. As shown in Tab. 2 (*SCP* column), we observe that the styled models perform consistently much better than their baseline counterpart for detection and pose estimation. When tested on the *CA* dataset, counter-intuitively, these models underperform in detection. One potential reason is that the network has never seen the complex vase dataset during training. Conversely for pose estimation, styled models unambiguously are better for both *SCP* and *CA* datasets. Specifically, styled models, which were not trained on *CA*, give a considerable jump in performance: 7.62 (mAP) & 8.06 (mAR) when tested on *CA*.

Tuned vs. Styled-Tuned models. (Results B, Tab. 3) With the goal of enhancing pose estimation on our *CA* dataset, a naive approach is to fine-tune on this data, we call these models as *Tuned models*. Then, we take the styled models (Tab. 2), which have already learned the styles of *CA* data, and fine-tune them on our *CA* data (*Styled-Tuned* or $Sd \rightarrow Td$). As seen in Tab. 3, the $Sd \rightarrow Td$ models give a better performance as compared to their *Tuned* counterparts for pose estimation. Irrespective of the combination of α and SS, the pose models tend to perform better. We argue that this is partly because the models gradually learn the styles (*SCP*), while optimising for the main task. During training, the *Styled* models (Tab. 2) are able to see the different spectrum of style intensities. They adapt the styles while maintaining a consistent performance over the main task. However, for person detection the performance of $Sd \rightarrow Td$ model is detrimental in comparison to the *Tuned* counterpart. One reason for this can be attributed to the overlapping objects: animals and persons – which makes the person detection more difficult for the $Sd \rightarrow Td$ models as compared to their *Tuned* counterparts. Another reason for lower precision is the lack of ground truth annotations for side characters of the scene. For poses however, the overlap of keypoints when compared to the bounding boxes is very small and hence the model generalizes better from styled models in comparison to directly tuned models.

Table 3. **Results B:** Comparing *tuned* model with the *styled-tuned* (Sd→Td) model, with different combinations of α (stylization factor) and SS (**RB** or **CA**), for (a) pose estimator (b) and detector. $\alpha = 0.5$ or $\alpha = U$ (randomly sampled from a uniform distribution: $U = \text{uniform}(0, 1)$). All values in *mAP*, except CA_{mAR} (*mAR*). The (+/-) is in reference to the corresponding baselines.

Model	CP	SCP	CA	CA_{mAR}	α	SS
Tuned	14.0	9.3	65.6	72.3	-	-
Sd→Td	11.8 (-2.2)	10.5 (+1.2)	66.8 (+1.2)	73.3	0.5	RB
	20.3 (+6.3)	14.5 (+5.2)	67.2 (+1.6)	73.3	<i>U</i>	RB
	34.9 (+20.9)	22.4 (+13.1)	66.6 (+1.0)	73.1	0.5	CA
	28.0 (+14.0)	18.5 (+9.2)	67.1 (+1.5)	73.6	<i>U</i>	CA

(a) Pose Estimation

Model	CP	SCP	CA	CA_{mAR}	α	SS
Tuned	-	-	49.4	37.0	-	-
Sd→Td	-	-	44.3 (-5.1)	32.9 (-4.1)	0.5	RB
	-	-	43.0 (-6.4)	32.6 (-4.4)	<i>U</i>	RB
	-	-	43.7 (-5.7)	33.4 (-3.6)	0.5	CA
	-	-	43.9 (-5.5)	32.9 (-4.1)	<i>U</i>	CA

(b) Person Detection

Table 4. **Results C:** Comparing *tuned* model with the *styled-tuned* (Sd→Td) model, by training on different quantities (25 %, 50 %, 75 %, 100 %) of the *CA* data for pose estimation. It clearly shows that the styled model learns quicker. All values in *mAP*, $\alpha = 0.5$ or $\alpha = U$ (randomly sampled from a uniform distribution: $U = \text{uniform}(0, 1)$), and SS = **RB** or **CA**

Model	25 %	50 %	75 %	100 %	α	SS
Tuned	61.3	65.0	65.1	65.6	-	-
Sd→Td	60.6	65.4	65.1	66.8	0.5	RB
	62.1	64.7	66.5	67.2	<i>U</i>	RB
	60.6	65.2	65.4	66.6	0.5	CA
	61.4	64.8	65.7	67.1	<i>U</i>	CA

With Tab. 2 and Tab. 3, we were able to enhance the performance of pose models, with *styled* as well as *styled-tuned* models. *Styled* models can help to improve the performance with a 7.7 pp (*mAP*) jump in performance, without any labels. While the *styled-tuned* models show that fine-tuning with *styled* models is generally beneficial for the performance, the *Sd→Td* model for pose gives a significant 1.6 pp (*mAP*) performance improvement for *CA* dataset over its *Tuned* counterpart.

Influence of data quantity. (Results C, Tab. 4): Tab. 4 shows that *Styled-Tuned* models learn faster than their *Tuned* counterpart, for each of the corresponding splits of *CA* data (model with $\alpha = U$, SS=RB is more consistent). Specifically, the model with $\alpha = 0.5$, SS=RB and 50% of *CA* data gives equivalent performance to the *Tuned* model trained with whole *CA* data. We see that the deep learning based models converge faster when they have a suitable initialization of weights [18]. We argue that training on styled data helps the model to get a better initialization with respect to the dataset distribution. Consequently, the convergence is faster.

Table 5. **Perceptual Loss Comparison:** Comparing different loss combinations using the *styled* or *styled-tuned* ($Sd \rightarrow Td$) model on the *CA* dataset for (a) pose estimation and (b) detection, *i. e.*, just the detector or pose loss L_{det}/L_{pose} or in combination with perceptual loss L_{comb} in two different variants (Eqs. (1a) and (1b)). For (a): $\alpha = 0.5$ and for (b): $\alpha = U$. *Style-Set* (*SS*) is *CA* for both.

Model	mAP	mAR	L_{comb}
Styled	30.6	36.9	L_{pose}
	30.5	36.5	L_{comb1}
	30.5	36.5	L_{comb2}
Sd→Td	66.6	73.1	L_{pose}
	67.2	73.6	L_{comb1}
	67.2	73.6	L_{comb2}

(a) Pose Estimation

Model	mAP	mAR	L_{comb}
Styled	6.5	8.6	L_{det}
	11.2	11.0	L_{comb1}
	9.8	11.2	L_{comb2}
Sd→Td	43.9	32.9	L_{det}
	45.4	34.7	L_{comb1}
	45.7	34.0	L_{comb2}

(b) Person Detection

Perceptual Loss Comparison: Tab. 5 shows the influence of perceptual loss for the model performance. L_{comb1} is the experiment with adaptively weighing the detector as well as pose losses (Eq. 1a). For L_{comb2} (Eq. 1b), we determined the values of λ s through a parameter search [1]: $\lambda_1 = 0.43$ and $\lambda_2 = 0.92$ for person detection and $\lambda_1 = 0.47$, $\lambda_2 = 0.018$ for pose estimation. We present results for the combination $SS=CA$ & $\alpha = U$ for the detector, and we chose $SS=CA$ & $\alpha = 0.5$ for pose estimator.

Tab. 5 shows that the perceptual loss (adaptive: L_{comb1} or parameterised one: L_{comb2}) indeed helps the *styled* as well as $Sd \rightarrow Td$ models to improve their performance, in general. From Tab. 5a, we can see that the results for L_{comb1} and L_{comb2} are equal, however we have to note that this is an empirical observation for different values of lambda. Additionally, we note that perceptual loss does not harm the *styled* model for pose estimation, but actually helps since the *styled* model is fine-tuned on *CA*.

5.3 Qualitative Pose Estimation Results

Tabs. 3 & 5 show that *styled-tuned* models are consistently giving a better performance than any other method. We visualise the predictions of these models for comparison of their performances. Fig. 6 shows four characters (*wrestler*, *fleeing*, *persecutor*, *bride*) and their pose predictions from each of our 5 proposed models.

As shown in Fig. 6a, the *baseline* model is the poorest in pose predictions. It is not able to detect majority of keypoints, confuses between the limbs if multiple characters are present and incorrectly predicts the keypoint locations.

Styled model is generally much better (Fig. 6b) than the *baseline* model (also Tab. 2). It is able to predict more keypoints and does not get confused if multiple characters are present. However, it is not able to predict all the visible keypoints and sometimes (Fig. 6b, *last row*) gives worse performance than even a *baseline* model.

Tuned, *styled-tuned* ($Sd \rightarrow Td$) and *styled-tuned* with perceptual loss ($Sd \rightarrow Td_p$) models are overall quite superior to *baseline* and *styled* models. They are able to predict almost all of the visible keypoints, do not confuse between multiple characters and are quite precise with the keypoint locations. However, there are subtle differences that make $Sd \rightarrow Td$ models better. They are able to predict all the visible keypoints as shown in Fig. 6d and Fig. 6e, whereas tuned models miss some (*e. g.*, Fig. 6c where the shoulder joints are missing). The keypoint location precision is also improved using $Sd \rightarrow Td$ models. Visually it is difficult to generalise if models with perceptual loss are better or not, however, they are more precise (Fig. 6e, third row ankle is corrected, but a shoulder is missed).

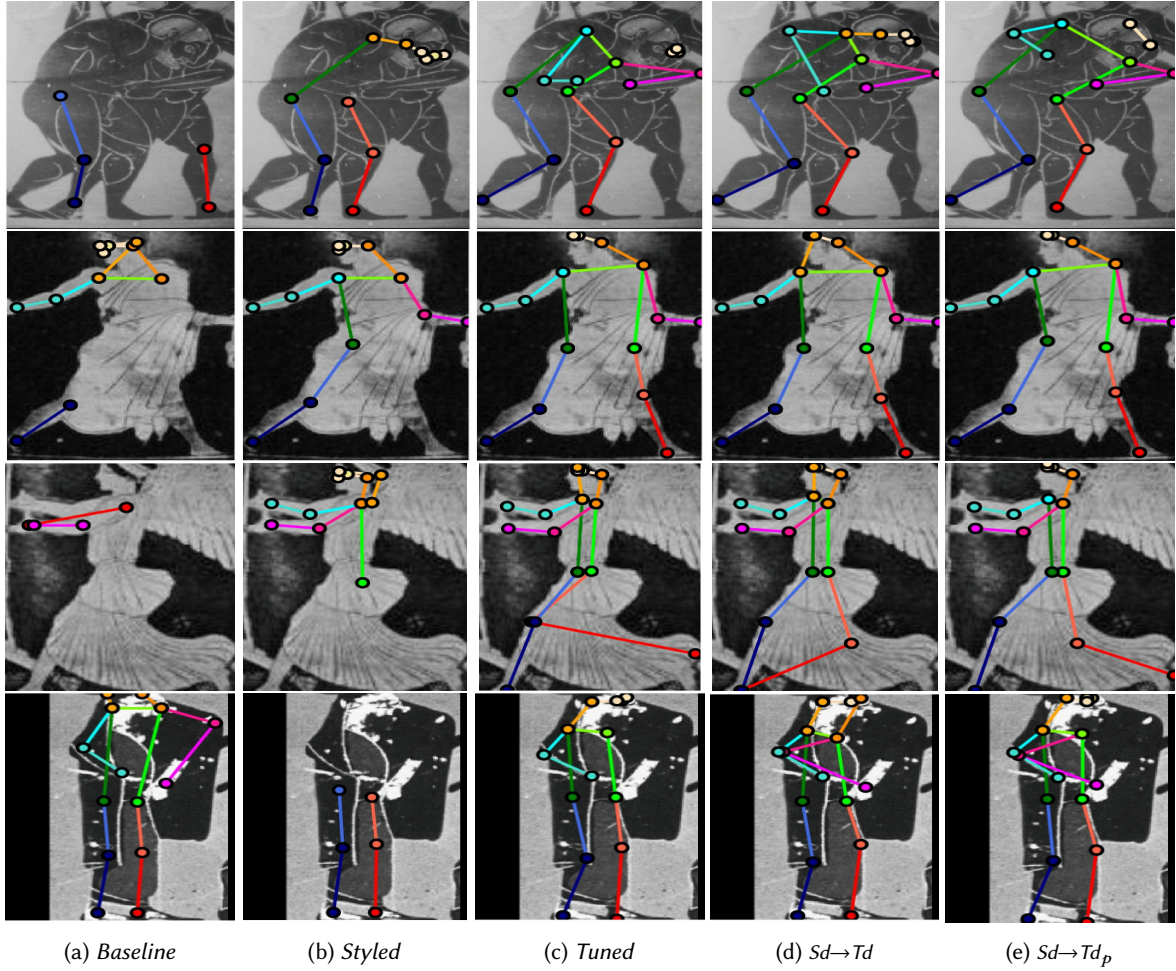


Fig. 6. **Pose models comparison:** Pose Predictions on 4 examples each from (a) *baseline*, (b) *styled*, (c) *tuned*, (d) *styled-tuned* ($S_t \rightarrow T_d$) and (e) *styled-tuned* with perceptual loss ($S_d \rightarrow T_{d_p}$) models. The results clearly show the superiority of predicted poses with the $S_t \rightarrow T_d$ and $S_d \rightarrow T_{d_p}$ models. The characters starting from the top are called *wrestler*, *fleeing*, *persecutor* and *bride*

6 POSE-BASED RETRIEVAL

Our experiments (Sec. 5) showed that *Styled* models and *Styled-Tuned* models achieve better keypoint detection results, quantitatively and qualitatively, than their corresponding counterparts. In this section, we show that our two-step training pipeline is also beneficial for discovering similar images based on character poses. We call the process of retrieving images based on poses as *pose-based retrieval*.

Table 6. **Retrieval Results:** The **(C)*** models show the retrieval values based on *characters*, where as the **(S)*** models show for the *scenes*. **P** is *Precision*, and **mAP** is *mean-Average Precision*. $\alpha = U$; **SS**: *Style-Set*. $Sd \rightarrow Td^*$ are *style-tuned* models, where $p1 = L_{comb1}$ (Eq. 1a) and $p2 = L_{comb2}$ (Eq. 1b)

Model	P@1	P@5	mAP	α	SS
(C) Baseline	31.7	25.5	21.5	-	-
(C) Styled	37.3	30.2	23.1	<i>U</i>	CA
(C) Tuned	43.0	39.8	27.5	-	-
(C) $Sd \rightarrow Td$	47.7	42.2	28.3	<i>U</i>	CA
(C) $Sd \rightarrow Td_{p1}$	45.7	41.4	28.0	0.5	CA
(C) $Sd \rightarrow Td_{p2}$	48.3	41.1	28.4	0.5	CA
(S) Baseline	43.6	43.2	35.9	-	-
(S) Styled	46.9	43.6	37.3	<i>U</i>	CA
(S) Tuned	56.4	52.7	41.5	-	-
(S) $Sd \rightarrow Td$	58.8	55.2	42.1	<i>U</i>	CA
(S) $Sd \rightarrow Td_{p1}$	57.8	53.5	41.5	0.5	CA
(S) $Sd \rightarrow Td_{p2}$	58.8	53.4	41.8	0.5	CA

6.1 Experimental Setup

The database for image retrieval and discovery is built from the *CA* validation dataset. The database consists of 303 images and their respective detected poses for best of *baseline*, *styled*, *tuned*, $Sd \rightarrow Td$ and $Sd \rightarrow Td_p$ (with perceptual loss). We perform two retrieval experiments based on the class label for each image, which is either a *character* or *scene*. There are 15 unique *characters* (**C**) and 5 *Scenes* (**S**). Given a query image, we rank the retrieved images based on the OKS metric [32]. In order to evaluate the retrieval method, we use the precision as: $P^* = \frac{TP^*}{TP^* + FP^*}$, where $*$ = @ k , consequently $P@k$:= Precision at k ; TP := true positives, FP := false positives and FN := false negatives. We report $P@k$ and mAP , for $k=1$ and $k=5$. In all our experiments, we exclude the self-retrieval (query itself) from the evaluation. For this task, we compare all the presented models to highlight the quality of our proposed models from an application perspective. The focus of this work is on enhancing poses for Greek vase paintings and not presenting a novel image retrieval method and hence we do not compare with SOTA image retrieval methods.

6.2 Retrieval Results and Discovery

Tab. 6 presents our pose-based retrieval results. We observe that *styled-tuned* models are consistently better for **C** and **S**, and the *styled* are better than *baselines* counterparts. Fig. 7 displays a query image (a) along with the top-5 ranked retrievals for the six different evaluated models. It can be observed that the tuned and the styled-tuned models outperform the baseline and the styled models. Fig. 7 row 1 shows poor retrieval results for the baseline model. For a pursuit scene, the first two retrieved samples belong to a wrestling scene while the last 2 belong to leading of the bride scene. The styled model (row 2) is already better wherein all the five retrievals belong to the pursuit scene, however at a character level, it retrieves a persecutor at the 5th retrieval. Tuned (row 3) and *styled-tuned* (row 4-6) models perform similarly well. On a closer look, we see that the 2nd and 4th retrieved samples of the *styled-tuned* model are closer to the query sample compared to the tuned model.

7 CONCLUSION

We presented a two-stage training approach for using style transfer and transfer learning in combination with perceptual consistency to improve pose estimation in ancient Greek vase paintings. We show that the use of styled transfer learning as a domain adaptation technique for such data significantly improves the performance

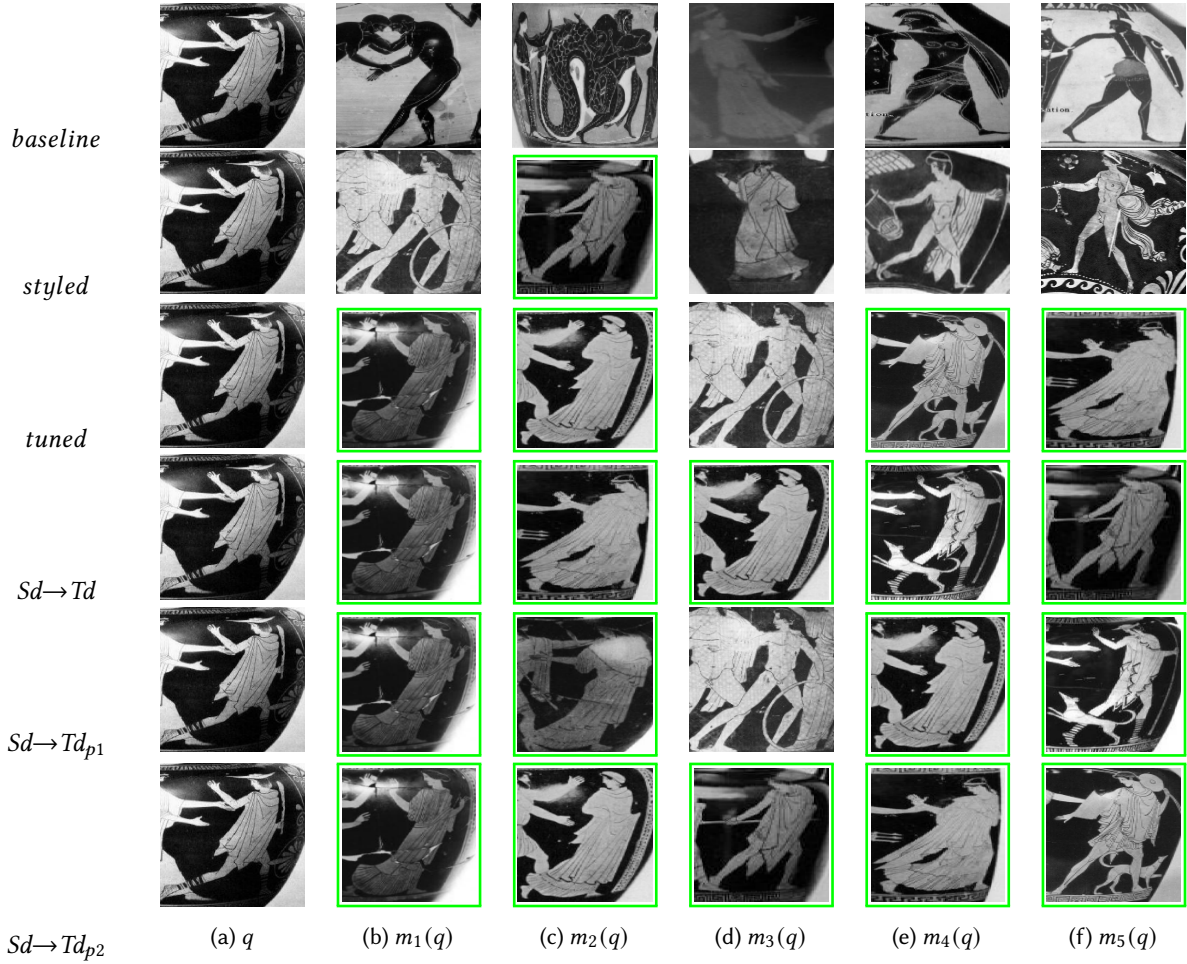


Fig. 7. **Discovery and Retrieval comparison:** (a) are query images (*fleeing character, pursuit scene*), and the remaining 5 columns ($m_i(q)$) are the five most similar images across four models; 1st row – *baseline*, 2nd row – *styled*, 3rd row – *Tuned*, 4th row – $Sd \rightarrow Td$, 5th and 6th row – $Sd \rightarrow Td_{p1}$ and $Sd \rightarrow Td_{p2}$ respectively. The results clearly show that the styled-tuned models retrieve the most precise results based on poses.

of state-of-the-art pose estimation models on unlabelled data by 6 % mean average precision (mAP) as well as mean average recall (mAR). We also analysed the impact of styles as progressive learning in a comprehensive manner showing that models learn generic domain styles. We experimentally showed that our proposed method outperforms their corresponding counterparts for human pose estimation. In general, our method can be applied to diverse unlabelled datasets without explicit supervised learning. Our method also provides a way for exploring diverse cross-domain datasets with low or no labels using human poses as a tool. Finally, we also show that our method can be used for pose-based image retrieval and discovery of similar, relevant poses and corresponding scenes in collections such as ancient Greek vase paintings. For future work, we plan to a) introduce the geometric structures of vases in COCO and Styled-COCO datasets as an augmentation technique during training, b) use shape information of the persons into our framework or using segmentation as a prior for vase paintings.

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